

Where injected work goes in a hard-disk box: a proof ladder for an energy-transfer experiment

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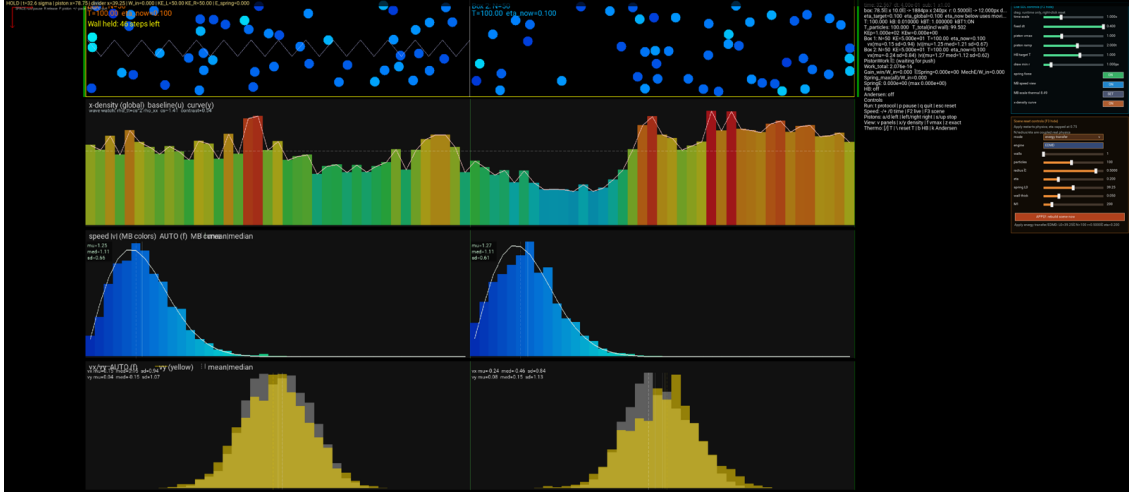
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Abstract

A piston compresses a hard-disk gas in a closed box. We ask what fraction of the injected work can still be captured as ordered energy, and we refuse to answer until each simpler statement below it is closed: the ledger, the quasi-static baseline, the finite-rate excess, the spring, the divider, the chain. Levels 0–2 are closed here. The headline of Level 2 is that a closed compartment has *no* linear friction: the force autocorrelation of a held piston integrates to zero at the sound traversal time, and the finite-rate cost is second order, $\langle W \rangle - W_{\text{qs}} = Au^2$, of which roughly half is energy the protocol itself launches.

1 The box and the ladder

Geometry A: two compartments of $N_s = 50$ disks either side of an immovable divider, a piston on the right. $k_B T = m = \sigma = 1$, $H = 10$, compartment 38.75σ , gas compression $\Delta x = 3.93\sigma$.



2 Level 0 — the ledger closes

$R_E = W - [\Delta KE_{\text{gas}} + \Delta KE_{\text{walls}} + \Delta E_{\text{spring}} + Q]$ and the matching momentum residual, from the per-event impulse log. **Measured:** $|R_E|/W \leq 8 \times 10^{-12}$, $|R_p|/\sum |\Delta p| \leq 1 \times 10^{-13}$ over all 75 pilot trajectories. **Passed.**

3 Level 1 — the quasi-static baseline

$W_{\text{qs}} = N_s k_B T (T_f/T_i - 1)$ with $\ln(T_f/T_i) = \int Z d \ln \eta$ along the path, Z measured in the finite box rather than taken from the bulk EOS. **Measured:** $W_{\text{qs}}^{\text{finite}} = 7.4685 \pm 0.0131 k_B T$; extrapolating the measured work to $u \rightarrow 0$ with the (quadratic) form gives $W(0) - W_{\text{qs}} = +0.0043 \pm 0.0158 k_B T$, 0.3σ . **Passed**, and on the physical side.

4 Level 2 — the finite-rate excess

4.1 Slow end

$$\zeta(\lambda) = \beta \int_0^\infty \langle \delta F(0) \delta F(t) \rangle dt, \quad W_{\text{diss}} \simeq \zeta u \Delta x. \quad (1)$$

Measured: the integral is not Markovian. It falls from 2.55 at zero lag — the Enskog collisional friction — and is cancelled by the returning sound wave at $L/c_s = 19.8\sigma$, confirmed at the compressed end of the path where the crossing moves to 18.0σ against a predicted 15.9. Its zero-frequency limit is the hydrodynamic viscous friction $\zeta_{\text{hyd}} = L_y \Gamma / X_p = 0.081 \pm 0.010$, which predicts a linear term of $0.32 \pm 0.04 u$ — below the resolution of 760 trajectories, and consistent with the measured slope -0.50 ± 0.67 for $u \leq 0.05$. Γ is taken from Paper 1's divider linewidth, not assumed.

What is there instead is quadratic: $A_{\text{step}} = 25.9 \pm 4.3$ against $A_{\text{ramp}} = 11.3 \pm 2.5$ when the piston is ramped over $2L/c_s$ instead of stepped (3.0σ apart). Roughly 14 of the 25 is energy the step protocol launches; what a gentle push leaves is consistent with the steady-flow value $N_s m/6 = 8.3$.

4.2 Fast end

$\langle W \rangle \simeq 2N_s m (\Delta x/L) u^2$, no free parameter. **Measured:** 0.986 of prediction at $u = 10$, $\Delta x = 7.08\sigma$. The apparent 8% excess at short travel is the contact layer: the strip a face can strike (width Δx starting at face $-r$) holds $1.13\times$ the uniform-density count at $\Delta x = 0.75\sigma$. Using the measured count the ratio becomes 0.96–1.03 with no trend.

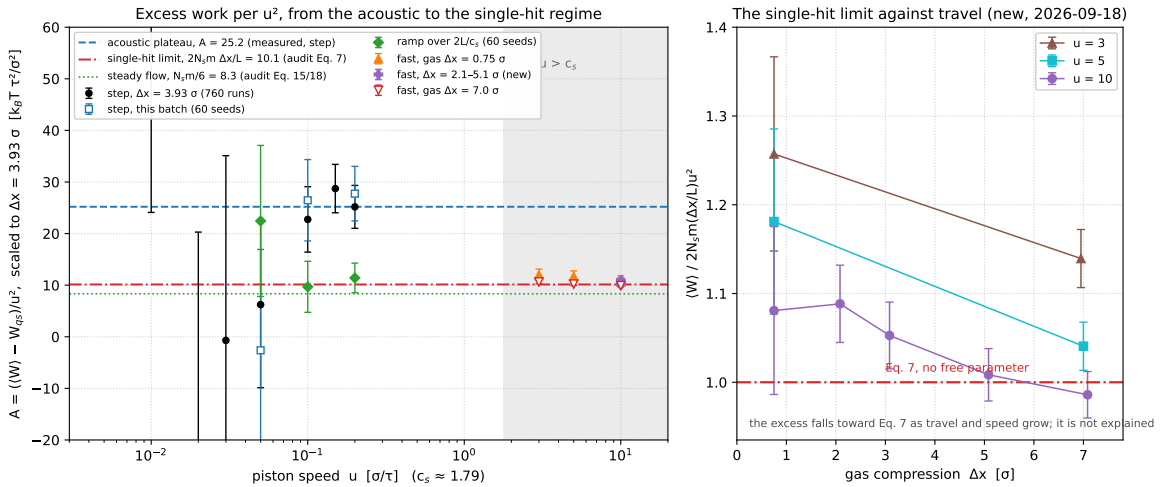


Figure 1: $A(u) = (\langle W \rangle - W_{\text{qs}})/u^2$ from the acoustic to the single-hit regime, every point scaled to a common travel, with the travel scan at right.

5 Level 3 — can a spring capture it?

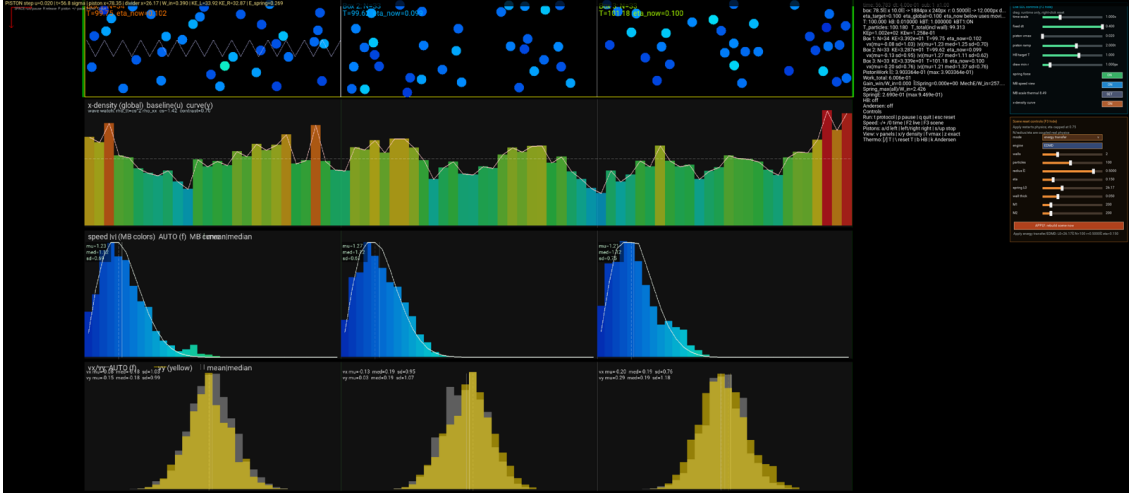
Not started. Geometry C: one gas drives a spring-loaded wall. **Pass criterion:** $E_{\text{spring,max}}$ agrees with the quasi-static prediction $\frac{1}{2}kx_{\text{eq}}^2$ when $\tau_{\text{push}} \gg T_w$ and with $\frac{k}{k+k_{\text{gas}}} \frac{\Delta p_w^2}{2M}$ when $\tau_{\text{push}} \ll T_w$, each within 2σ .

6 Level 4 — transmission through a free divider

Not started. Fast pressure channel against slow heat channel; $\mathcal{T}(t) = \Delta K E_{\text{gas}2}(t)/W_{\text{in}}$, with $\mathcal{T}_{\text{slow}} \rightarrow 1/2$ for symmetric gases.

7 Level 5 — the chain

Not started. $\epsilon = E_{\text{spring,max}}/W_{\text{in}}$ against piston speed, divider mass and the number of walls; the prediction is the *ordering*, geometric intermediate masses beating badly matched ones.



8 Level 6 — information

Not started, and gated on a canonical seeder. $\beta \langle W_{\text{diss}} \rangle \geq I_{\text{mem}} - I_{\text{pred}}$. Note that the Jarzynski benchmark must be the finite-box ΔF , exactly as Level 1 needed the finite-box W_{qs} .

Reproducibility

Raw trajectories are not in the repository; each cell carries `summary.csv` and `00_COMMAND.md`, and Level 2 an 85 kB `stop_digest.csv` from which the entire analysis reproduces. Batch scripts are in `../experiments/run_scripts/`.